



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
THAPATHALI CAMPUS**

Minor Project Proposal

on

**A TERRAIN-BASED GEOLOCATION APPROACH USING MOUNTAIN SKYLINES
AND DIGITAL ELEVATION MODELS IN NEPAL**

Submitted By:

Bibha Ray	(THA080BCT009)
Prasiddha Raj Poudel	(THA080BCT027)
Reema Kasula	(THA080BCT032)

Submitted To:

Department of Electronics and Computer Engineering
Thapathali Campus
Kathmandu, Nepal

June 2026

ABSTRACT

This project explores a terrain-based localization system designed for the rugged Himalayan landscape of Nepal, where GPS reliability can be affected by signal multipath and ground-level visual data is often limited. The system utilizes Global 30m Elevation Data (Copernicus) (GLO-30) Digital Elevation Model (DEM) data to generate a searchable database of synthetic horizon profiles. For image processing, an efficient segmentation architecture, such as MobileNet-V3-UNet, is proposed to isolate the skyline from user-captured photos. To address potential orientation uncertainty, the project investigates matching strategies—possibly involving Euclidean distance for initial pruning and Dynamic Time Warping (DTW) for refining matches—to account for perspective variations. The system can be evaluated using georeferenced data to assess localization potential and computational feasibility in high-relief environments.

Keywords—DEM, Skyline Localization, Himalayan Topography, Computer Vision, MobileNet-V3, DTW.

1 INTRODUCTION

1.1 Background and Motivation

In high-altitude regions such as the Nepal Himalayas, satellite-based navigation often encounters challenges due to signal multipath errors and occlusion caused by steep topographic relief [1, 2]. Conventional visual localization services frequently have limited coverage in these remote environments because they rely on ground-level imagery databases that are predominantly urban-centric [3, 4]. However, prominent mountain silhouettes can provide a stable, long-range topographic reference that remains temporally consistent across seasonal changes [5, 6]. Terrain-based localization attempts to leverage these silhouettes by comparing features extracted from user-captured photographs against synthetic horizon data derived from DEMs [3, 7]. This project is motivated by the potential for a robust localization approach that can function in regions where standard GNSS signals are unreliable.

1.2 Problem Definition

The primary objective is to estimate geographic coordinates and orientation based on the visible skyline. This involves several key challenges:

1. **Feature Extraction:** Isolating the terrain-sky boundary in photographs under varying atmospheric and lighting conditions.
2. **Data Representation:** Organizing synthetic horizon data in a format suitable for efficient storage and retrieval across target areas.
3. **Search and Retrieval:** Developing a matching logic capable of identifying the most likely candidates from a reference index, even when camera orientation is unknown.

1.3 Objectives

- To develop a pipeline for generating synthetic horizon data from DEM sources and deploy a computer vision architecture for robust skyline segmentation.
- To implement a matching strategy that utilizes similarity metrics for localization and evaluate the system using georeferenced imagery within selected high-relief regions.

1.4 Scope

The project focuses on terrain-based localization in mountainous areas of Nepal. The system is intended to process static images where a portion of the horizon is visible. The scope is limited to the primary skyline boundary and does not currently focus on internal ridge-lines or scenarios where the skyline is entirely occluded by weather or

local obstacles. The project aims to demonstrate feasibility within specified test sites rather than providing a global or nationwide deployment.

2 LITERATURE REVIEW

2.1 Digital Elevation Models (DEMs) for Terrain Representation

Digital Elevation Models serve as the topographic foundation for terrain-based localization, representing surface relief through gridded elevation values [8, 9]. Historically, the Shuttle Radar Topography Mission (SRTM) and ASTER GDEM provided the baseline for global coverage, but they often struggle with vertical inaccuracies in high-relief environments [8]. Recent assessments indicate that Copernicus GLO-30 consistently outperforms these older models in mountainous regions, offering superior precision with a Root Mean Square Error (RMSE) of approximately 2.34m in high-relief sites [9]. While 10m resolution data is often viewed as a common baseline for urban or low-relief localization, research suggests that for the massive vertical scale of Himalayan peaks, 30m resolution is sufficiently representative [8, 9]. This resolution offers a strategic balance, providing high-fidelity topographic fingerprints while significantly reducing the database size and computational overhead required for large-scale regional processing in Nepal.

2.2 Horizon and Skyline-Based Geolocation

Skyline matching identifies geographic coordinates by comparing silhouettes extracted from photographs with synthetic profiles rendered from a DEM [3, 4]. This approach is typically treated as a retrieval task against a pre-computed database of horizons, which is essential for navigation in environments where GPS is unreliable or ground-level imagery like Google Street View is sparse [3, 6]. A significant limitation in existing work is the sensitivity to “bland” or feature-poor skylines, which can lead to localization failure [3]. To combat this, the literature emphasizes the need for dense sampling of candidate viewpoints. The use of metric coordinate systems (e.g., UTM) for grid generation is recognized as a best practice to ensure uniform distance across rugged terrain, preventing the spatial distortions inherent in standard geographic coordinates when calculating line-of-sight in mountainous areas.

2.3 Viewshed Analysis and Profile Generation

Generating a reference database requires viewshed analysis to compute the visible horizon from a grid of candidate locations using ray-casting algorithms [7, 10]. The computational cost of ray-casting is a function of the search radius and the angular resolution of the profile. Standard practices involve checking lines of sight up to 50km to capture distant high-altitude peaks that define the global horizon [3, 10]. While high-resolution profiles (e.g., 0.5 degrees) provide more detail, they increase the memory footprint of the searchable index. Efficient storage and retrieval are often managed using

numerical fingerprints in optimized structures like NumPy arrays or FAISS (Facebook AI Similarity Search) indices, which allow for rapid querying across thousands of potential geographic candidates.

2.4 Computer Vision for Horizon Extraction

Automatic extraction of the skyline from raw images is a prerequisite for localization [5, 11]. Traditional methods utilizing Canny edge detection and gradient maps frequently fail under the haze, low-contrast, and cloudy conditions common in high-altitude photography [3, 11]. Modern deep learning architectures offer increased robustness. Specifically, the MobileNet-V3-UNet architecture provides an efficient compromise between the high-pixel accuracy of traditional UNets and the low-latency requirements of mobile-ready systems [11]. Despite the efficacy of deep learning, these networks require large datasets. In regions like the Himalayas where labeled ground truth is scarce, the literature supports a “synthetic-to-real” approach, where DEMs are used to render thousands of synthetic silhouettes for training, supplemented by simulated noise, clouds, and textures to improve model generalization.

2.5 Similarity Matching and Orientation Estimation

A major challenge in mountain geolocalization is the lack of accurate compass (heading) metadata. While astronomical cues like sun position (via ephemeris tools) can constrain the search space, they are weather-dependent [3, 4]. Consequently, the literature favors a rotational “sliding window” approach, comparing the extracted horizon against all possible orientations in the database [4]. For the matching process, a tiered strategy is often employed to maintain scalability: a fast initial search (Euclidean or Cosine similarity) identifies a subset of top candidates, which are then refined using Dynamic Time Warping (DTW) [6]. DTW is particularly valued for its ability to handle “time-axis” distortions caused by lens zoom or atmospheric warping. To keep the compute time linear, techniques such as Early Abandoning and Pruning (EAP) are critical for making DTW viable for real-time applications.

2.6 Validation via Ground-Truth Datasets

Validating terrain-based localization systems requires datasets with verified GPS and heading metadata. In the absence of massive hand-labeled datasets for the Nepal region, researchers frequently utilize Google Street View photospheres [3]. These provide a globally consistent ground-truth set that allows for the validation of both the horizon extraction accuracy and the similarity matching logic. This benchmarking is essential to ensure that the transition from synthetic training data to real-world photography does not result in a significant drop in localization precision.

2.7 Conclusion and Project Gaps

The existing research identifies several critical gaps: (i) the high computational cost of matching silhouettes without orientation data, (ii) the scarcity of labeled training data for the Himalayas, and (iii) the difficulty of maintaining accuracy in hazy conditions. This project addresses these gaps by implementing an efficient localization pipeline for Nepal that utilizes Copernicus GLO-30 DEM for topographic precision, a MobileNet-V3-UNet for robust haze-resistant segmentation, and a two-step similarity search (Euclidean + DTW) to ensure scalable, orientation-independent matching across the rugged Himalayan landscape.

3 SYSTEM ARCHITECTURE AND METHODOLOGY

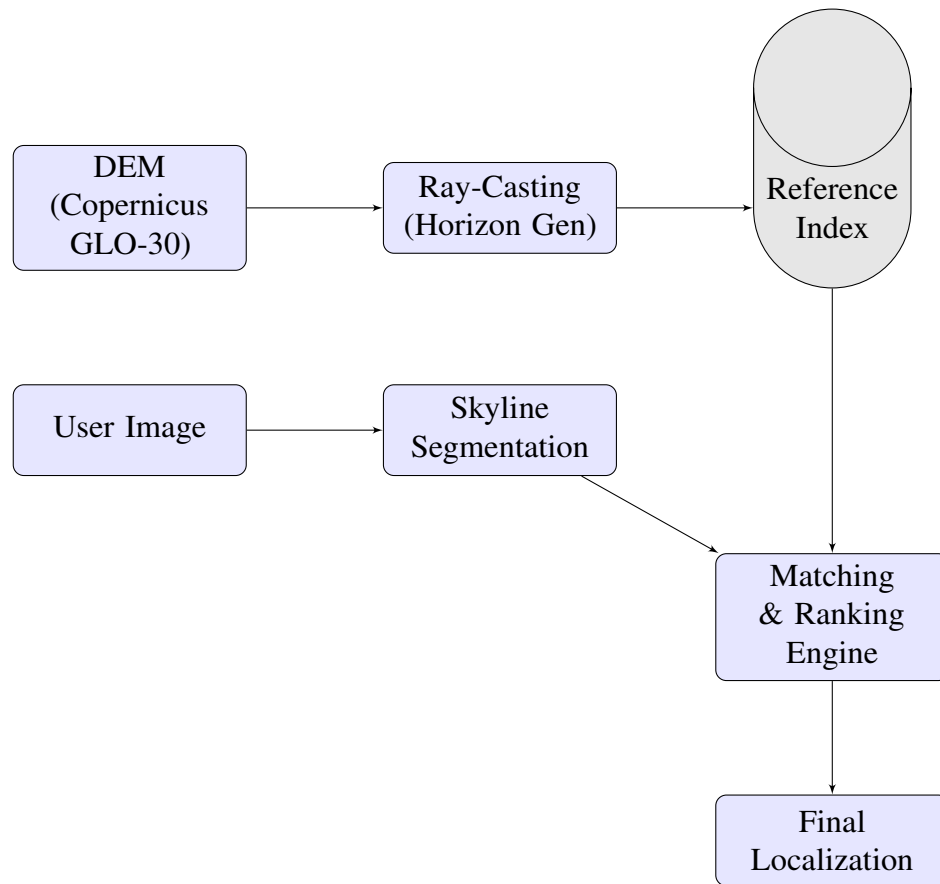


Figure 3-1 System Block Diagram showing the interaction between data generation and image processing.

The accuracy of the system is grounded in the Digital Elevation Model (DEM) used to represent the topography. We plan to utilize Copernicus GLO-30 (30m) data. While 10m data is a common baseline, research indicates that for massive Himalayan peaks, 30m resolution is sufficiently representative and offers substantial benefits in reduced database size and faster processing.

To build the search space, the region is sampled using a grid of candidate viewpoints. A baseline grid of 500-meter spacing is proposed for specific target regions like Khumbu or Annapurna. These points are mapped using a metric coordinate system like UTM Zone 45N to ensure uniform distance across rugged terrain, though standard WGS84 coordinates may be used for final storage and mapping to maintain compatibility with GPS data.

From every point in the grid, a 360-degree horizon profile is generated. This is achieved through ray-casting, where the elevation angle of the skyline is calculated at set intervals—typically every 1 degree (360 values), though 0.5 degrees (720 values) remains an option if the memory budget allows. This process involves checking the

line of sight for up to 50 kilometers. The result is a numerical fingerprint stored in a searchable index; for the scale of a single region, a simple NumPy array or a lightweight FAISS index will serve as the permanent reference.

The primary goal is to isolate the terrain-sky boundary efficiently. We are starting with a MobileNet-V3-UNet architecture. Since large, hand-labeled datasets for Nepal are rare, the DEM itself will be used to render thousands of synthetic silhouettes for training. These synthetic images will include simulated noise—such as varying sky conditions, clouds, and ground textures—to help the model generalize. If the UNet struggles with boundary fidelity or becomes a bottleneck, a Canny edge-detection heuristic will probably be used.

A major challenge is determining the camera’s compass heading. While identifying the sun’s glare using ephemeris tools like Pysolar is a potential improvement for narrowing the search, it is highly dependent on clear weather and accurate timestamps. Instead, the system will prioritize a sliding window approach, comparing the photo’s horizon against every possible rotation in the database. This ensures the system remains functional even when no astronomical data is available.

A similarity search is used to pull the most likely candidates from the database. We will use a two-step matching process: first, a fast Euclidean or Cosine similarity search to find the top 50–100 candidates, followed by Dynamic Time Warping (DTW) for final ranking. We will prioritize Early Abandoning and Pruning (EAP) to keep compute time linear, ensuring peaks are matched even if they are slightly distorted by lens zoom or atmospheric haze. While the current implementation focuses on the primary visible skyline, this framework establishes a foundation for future development; subsequent iterations could be extended to utilize internal mountain ridge-lines to maintain localization in scenarios where fog or heavy haze occludes the distant global horizon.

Although RANSAC-based optimization to reach sub-grid accuracy is a possibility, it may not be viable within the current timeframe. We will validate the logic using Google Street View photospheres, which provide a ground-truth dataset with verified GPS and heading metadata to test the database.

4 EXPECTED OUTCOMES

The project is expected to result in a system that can take a user image and output a predicted geographic location and camera heading. The output will include a visual of the detected horizon line compared against the synthetic mountain shape from the database to show how the match was made. By providing a ranked list of the top candidate matches rather than a single point, the system can still be considered successful if the true location appears within the top-tier results, even when a perfect coordinate match is not achieved due to terrain similarities or image noise.

5 FEASIBILITY ANALYSIS

5.1 Technical Feasibility

The project's technical viability is supported by the availability of mature, well-defined software libraries. For processing GLO-30 DEM data and performing coordinate transformations, standard geospatial packages such as Rasterio or GDAL can be utilized. The implementation of the DTW logic and similarity search can leverage optimized libraries like SciPy or Facebook AI Similarity Search (FAISS), which are designed for efficient numerical computation. For the skyline segmentation task, deep learning frameworks such as PyTorch or TensorFlow may be employed to deploy a MobileNet-V3-UNet architecture. By utilizing these existing tools, the development can focus on the specific logic of terrain matching rather than low-level data handling or basic algorithm implementation.

5.2 Scope and Database Feasibility

To maintain a manageable computational and storage footprint, the system does not attempt to index a large geographic region. Instead, the project can be limited to specific Regions of Interest (ROIs), such as the Khumbu or Annapurna regions of Nepal. The database size may be further controlled through strategic constraints:

1. **Grid Sampling:** Viewpoints can be sampled at a coarse interval (e.g., 500m) rather than high-density 30m steps, significantly reducing the total number of entries in the index.
2. **Search Radius:** The ray-casting process for horizon generation can be capped at some radius like 50km. This prevents the inclusion of distant, irrelevant terrain data, keeping the memory requirements within the limits of standard consumer hardware.

5.3 Operational Feasibility

By focusing on a limited geographic scope and using a tiered search strategy, the system remains operationally viable. The matching engine might first employ a fast search e.g. Euclidean distance search to prune the database to a small subset of candidates (e.g., the top 50 matches). Only this subset would then be processed using the more computationally intensive DTW algorithm. This tiered approach ensures that localization can be achieved in a reasonable timeframe without requiring high-performance computing clusters, making the project feasible for execution on typical devices.

REFERENCES

- [1] B. Liu, P. Jia, W. Wang, D. Xu, J. Cheng, J. Dong, X. Han, Z. Zhao, C. Zhang, F. Hong *et al.*, “Enhancing cross-view uav geolocalization via lvlm-driven relational modeling,” *arXiv:2603.08063*, 2026.
- [2] A. Durgam, S. Paheding, V. Dhiman, and V. Devabhaktuni, “Cross-view geo-localization: A survey,” *IEEE Access*, vol. 12, pp. 192 028–192 050, 2024.
- [3] G. Baatz, O. Saurer, K. Köser, and M. Pollefeys, “Large scale visual geo-localization of images in mountainous terrain,” in *Proceedings of the 12th European Conference on Computer Vision (ECCV)*, 2012, pp. 517–530.
- [4] Y. Chen, G. Qian, K. H. Gunda, H. Gupta, and K. Shafique, “Camera geolocation from mountain images,” in *Proceedings of the 18th International Conference on Information Fusion (Fusion)*, 2015, pp. 1587–1596.
- [5] A. Norinder and H. Sandblom, “Panoramic skyline matching for geo-localization in archipelagic environments,” Master’s thesis, Lund University, 2025.
- [6] E. Villanueva Rourera, “Skyline matching: absolute localisation for planetary exploration rovers,” Master’s thesis, Institut Supérieur de l’Aéronautique et de l’Espace (ISAE-SUPAERO), 2022.
- [7] E. Tzeng, A. Zhai, M. Clements, R. Townshend, and A. Zakhor, “User-driven geolocation of untagged desert imagery using digital elevation models,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2013, pp. 237–244.
- [8] B. Purinton and B. Bookhagen, “Validation of digital elevation models (dems) and comparison of geomorphic metrics on the southern central andean plateau,” *Earth Surface Dynamics*, vol. 5, pp. 211–237, 2017.
- [9] C. J. Okolie, J. Mills, A. Adeleke, M. Smit, M. Peppia, A. Altunel, and I. Arungwa, “Assessment of the global copernicus, nasadem, aster and aw3d digital elevation models in central and southern africa,” *Geo-spatial Information Science*, vol. 27, no. 4, pp. 1362–1390, 2024.
- [10] C. R. Steger, B. Steger, and C. Schär, “Horayzon v1.2: an efficient and flexible ray-tracing algorithm to compute horizon and sky view factor,” *Geoscientific Model Development*, vol. 15, no. 17, pp. 6817–6840, 2022.
- [11] G. Yang, M. Wang, Q. Zhou, and J. Li, “Yunet: Improved yolov11 network for skyline detection,” *arXiv:2502.12449*, 2025.